

Ultimate Basic — C64 compiler

A modern BASIC-like language that compiles directly to 6502 machine code for the Commodore 64. Output: `.prg` files (VICE or real hardware) and `.d64` disk images.

Quick start

```
cargo build --release

# Compile to .prg (memory map printed on success)
ub build demo.ub -o demo.prg

# Verbose: also prints zero-page layout and hex dump
ub build demo.ub -v

# Compile + create .d64 disk image
ub build demo.ub --d64 disk.d64
```

Language reference

Variables and constants

```
var x = 10           # 8-bit integer (default)
var ptr: word = $0400 # 16-bit – two zero-page bytes, usable as 16-bit address
var f: float = 3.5    # Q8.8 fixed-point – hi byte = integer, lo byte =
fraction
var msg = "HELLO"     # string variable (pointer to inline PETSCII data)
var s: string = "TEXT" # string with explicit type
var scores = array(10) # byte array, 10 elements stored at $C000+
var times = array_word(8) # word array, 8 word elements stored at $C000+
const BORDER_ADDR = $D020 # compile-time constant (substituted inline, no ZP slot)
```

Keywords and identifiers are **case-insensitive**: `PRINT`, `Print`, and `print` are all valid.

Type	Width	Notes
<code>int</code>	8-bit	default for numeric literals
<code>word</code>	16-bit	two ZP bytes; can be used as address in <code>poke/peek</code>
<code>float</code>	16-bit Q8.8	hi byte = integer part (0–255), lo byte = fractional part
<code>string</code>	pointer	ZP pair → null-terminated PETSCII in code segment
<code>array(N)</code>	N bytes	byte elements; lives at <code>\$C000+</code> , not in ZP
<code>array_word(N)</code>	N×2 bytes	word (16-bit) elements; lives at <code>\$C000+</code> , not in ZP

Comments

```
# hash comment
rem this is also a comment
var x = 5 ; inline semicolon comment
var x = 5 : var y = 6 # colon separates statements on one line
```

Operators

```
x = x + 1
y = a * b - c / 2
z = x and 15          # bitwise AND
w = a or b            # bitwise OR
v = a xor b           # bitwise XOR
m = x shl 3           # shift left
n = x shr 2           # shift right
r = x mod 40          # 8-bit modulo (remainder); SEC/SBC/BCS loop
```

Comparisons: == != < > <= >= (return 1/0)

Increment / Decrement

```
inc x                # x = x + 1 (INC zp – single instruction)
dec x                # x = x - 1 (DEC zp – single instruction)
```

For word variables carry is handled: `inc` uses `INC lo; BNE skip; INC hi`; `dec` uses `LDA lo; BNE skip; DEC hi; DEC lo`.

Compound Assignments

```
x += 5               # x = x + 5
x -= 3               # x = x - 3
x *= 2               # x = x * 2
x /= 4               # x = x / 4
x and= 15            # x = x and 15 (bitwise AND)
x or= 64             # x = x or 64 (bitwise OR)
x xor= 255           # x = x xor 255 (bitwise XOR)
x shl= 2             # x = x shl 2
x shr= 1             # x = x shr 1
```

Print

```

print "HELLO"
print x
print x, y, "text"
print                                     # blank line

print spc(5)                            # print 5 space characters
print tab(20), "VALUE"                  # move cursor to column 20, then print
print "A", spc(3), "B"                  # mix freely

print "A=" + a                          # string + numeric var
print s1 + s2                           # two string vars → sequential print
print "Hello " + "World"                # two literals → compile-time fold
print chr$(13)                          # print by PETSCII code

```

chr\$

```

print chr$(65)                          # output character with PETSCII code 65 ('A')
var c = chr$(n)                          # store byte value n in variable c
print ">" + chr$(42)                     # usable in string concat

```

Branching

```

if x == 1 then
  print "YES"
else
  print "NO"
end

```

Select / Case

```

select x
case 1:
  print "ONE"
case 2:
  print "TWO"
else:
  print "OTHER"
end

```

`select expr` evaluates the expression once and compares it against each `case` value in order. The first matching case body is executed and control jumps to after `end`. The optional `else:` body runs if no case matches. All values must be 8-bit (0–255).

Loops

```

loop 5                # counted loop (5 iterations)
  print "HI"
end

loop                  # infinite loop
  x = x + 1
  if x == 5 then continue end # skip to next iteration
  if x == 100 then break end
end

for i = 1 to 10        # for..next (preferred)
  if i == 5 then continue end # skip to increment step
  print i
next

for i = 0 to 20 step 2
  print i
next i                 # variable name after 'next' is optional

loop i = 1 to 10       # legacy loop..end syntax – identical code
  print i
end

while x < 100
  x = x + 1
end

repeat                # do-while: body runs at least once
  x = x + 1
until x == 100        # exits when condition is true

```

Labels and goto

```

label main_loop
  x = x + 1
  if x < 10 then goto main_loop end

```

Forward `goto` (label defined later) is fully supported.

Subroutines

```

sub greet()
  print "HELLO!"
end

sub set_color(col)
  color border col
  color text col

```

```

end

greet()           # call with parens
call greet        # call keyword (no parens)
set_color(6)

```

Parameters are passed via dedicated zero-page slots. No recursion (slots are static).

Arrays

```

var scores = array(8)    # 8 bytes at $C000

scores[0] = 100           # constant index → STA $C000
scores[i] = 99            # variable index → STA (ptr),Y
var v = scores[i]         # LDA (ptr),Y
print scores[2]           # usable inline in print

var times = array_word(8) # 16 bytes (8×2) at $C000+

times[0] = $1234          # constant index → STA $C000 (lo), STA $C001 (hi)
times[i] = $5678          # variable index → ASL A for stride; (ptr),Y × 2
var t: word = times[1]    # LDA $C002, LDA $C003

```

16-bit (word) variables

```

var ptr: word = $0400    # two ZP bytes: lo=$00 hi=$04
poke ptr, 6              # STA (ptr),Y
var v = peek(ptr)        # LDA (ptr),Y

```

Bitmap graphics

```

graphics on              # VIC-II hires bitmap mode (320×200, 1bpp); bitmap at
$2000
graphics on multi        # VIC-II multicolor bitmap mode (160×200, 2bpp, 4
colours/cell)
graphics off             # return to text mode

gcls                     # clear bitmap (fills $2000-$3FFF) + set video matrix
colors

plot x, y                # set pixel at (x, y); x: 0-319, y: 0-199
plot erase x, y           # clear pixel (AND ~mask)
plot xor x, y            # toggle pixel (EOR mask) – flicker-free animation
circle x, y, r           # midpoint circle centered at (x, y) with radius r; clips
off-screen points
line x1, y1, x2, y2      # Bresenham line from (x1,y1) to (x2,y2); x: 0-255, y: 0-
199

```

```

paint x, y          # 4-connected flood fill from (x, y); fills clear pixels
                    # bounded by set ones

```

Both `graphics on` variants blank the display (`LDA $D011 / AND #$EF / STA $D011`) while switching VIC registers, then re-enable it in the target mode — prevents mode-switch glitches.

Screen and color

```

cls                # clear screen (KERNAL $E544)
cls fast           # fast fill: screen RAM + color RAM + HOME

color text 14       # text color register $0286
color border 6      # $D020
color bg 0          # $D021

screen 0, 0, 65     # write char 65 ('A') to screen RAM at col 0, row 0
($0400)
screen 10, 5, ch     # col 10, row 5 – col/row can be variables
screen 5, 3, 42, 7   # char 42 at col 5, row 3, color 7 (writes color RAM
$D800 too)
screen x, y, ch, col # all four arguments as variables

display on         # re-enable VIC display ($D011 DEN bit)
display off        # blank display

cursor 20, 10       # move cursor to col 20, row 10 (KERNAL PLOT $FFF0)
cursor x, y         # column from variable x (0-39), row from y (0-24)

print at 20, 10, "HELLO" # cursor(20,10) + print in one statement (no trailing
newline)
print at x, y, "Score:", score # any mix of exprs
print at 0, 0       # position only (no text, no newline)

scroll x 3          # set horizontal fine scroll: $D016 bits 0-2 = 3 (0-7)
scroll y 2          # set vertical fine scroll:  $D011 bits 0-2 = 2 (0-7)
scroll x n          # value can be a variable or expression (masked to bits
0-2)

```

`scroll x n` writes $(n \text{ AND } 7)$ into bits 0-2 of `$D016` (preserving bits 3-7). `scroll y n` writes $(n \text{ AND } 7)$ into bits 0-2 of `$D011` (preserving bits 3-7). Useful for smooth hardware scrolling: decrement each frame from 7 down to 0, shift screen RAM, reset to 7.

`screen col, row, char [, color]` writes directly to screen RAM ($\$0400 + \text{row} * 40 + \text{col}$) and optionally to color RAM ($\$D800 + \text{row} * 40 + \text{col}$). Constant col/row: address computed at compile time.

Ultimate 64 — CPU Speed

```

speed 4            # set CPU to 4 MHz (reads $D031, updates bits 0-3, writes
back)

```

```

speed 48      # 48 MHz (maximum speed on U64)
speed max     # same as speed 48 (alias)
speed off     # back to 1 MHz (alias for speed 1)

badlines on   # enable badline timing ($D031 bit 7 = 0, default C64
behaviour)
badlines off  # disable badline timing ($D031 bit 7 = 1, more CPU cycles)

var t = turbo() # 1 if turbo is active (bits 0-3 of $D031 != 0), 0 if at 1
MHz

```

Available MHz values: 1, 2, 3, 4, 5, 6, 8, 10, 12, 14, 16, 20, 24, 32, 40, 48. Constant values are rounded down to the nearest available speed at compile time. Variable values are treated as a raw speed index (0–15) and OR'd into bits 0-3 of `$D031`.

\$D031 index	MHz (U64)	MHz (U64 Elite-II)
0	1	1
3	4	4
6	8	10
11	20	24
15	48	64

Requires **U64 Turbo Control** set to **U64 Turbo Registers** or **Turbo Enable Bit** in the U64 config menu. On a regular C64 or emulator without the register, the `poke` to `$D031` is silently ignored.

Keyboard

```

var key = getch()      # busy-wait on $FFE4 until key; returns PETSCII code
var k   = inkey()      # non-blocking: returns PETSCII code, or 0 if no key
pressed
var j = joy(2)          # read joystick port 2; returns inverted bits 0-4
var j = joy(1)          # read joystick port 1
                        # bit0=up(1) bit1=down(2) bit2=left(4) bit3=right(8)

bit4=fire(16)
var mx = mouse_x()      # 1351 mouse X: SID POT X ($D419), 0-255
var my = mouse_y()      # 1351 mouse Y: SID POT Y ($D41A), 0-255
var mb = mouse_btn()    # mouse buttons: bit0=left (fire), bit1=right

```

Exit

```

bye          # JSR $E544 (clear screen), clear STOP flag, RTS to BASIC
exit         # alias for bye

```

Timing

```
wait 50          # wait 50 raster-line transitions (~3.2 ms)
wait raster 100  # spin until $D012 == 100 (raster-split effects)
delay 1          # wait 1 PAL frame (1/50 s ≈ 20 ms)
delay 20         # wait 20 frames ≈ 0.4 s; n can be a variable (0-255)
```

`delay N` counts N complete PAL frames using raster line 200 as the frame boundary.

SID Sound

```
sound 0, $1CAD, 25  # voice 0, freq $1CAD (≈ middle C PAL), 25 frames
duration
sound 1, freq_word, 50 # voice 1, freq from word var, 50 frames (1 s at 50 Hz)
sound 2, 0, 0        # voice 2, silence

sid volume 15        # master volume full ($D418 = $0F); range 0-15
sid volume 0         # silence (master volume = 0)
sid stop             # zero all 25 SID registers ($D400-$D418) – complete
silence
```

`sound <channel>, <freq>, <duration>` — duration in PAL frames (1/50 s each). Fixed ADSR: attack/decay \$09, sustain/release \$F0, sawtooth waveform. Master volume \$D418 always set to \$0F.

`sid volume N` writes N to \$D418. Bits 0-3 = volume (0-15), bits 4-7 = filter mode. `sid stop` emits a 10-byte zero-fill loop — faster than 25 individual pokes.

Sprites

```
sprite 0, x, y, $2000 # sprite 0: set X, Y position and data pointer
sprite 0, x, y        # without data pointer (keeps existing)
sprite on 0           # enable sprite 0 ($D015 |= bit0)
sprite off 0          # disable sprite 0 ($D015 &= ~bit0)
sprite color 0, 7     # sprite 0 color = yellow ($D027)
sprite multicolor 0, on # enable multicolor mode for sprite 0 ($D01C |= bit0)
sprite multicolor 0, off # disable multicolor mode ($D01C &= ~bit0)
sprite expand x 0, on  # double width ($D01D |= bit0)
sprite expand x 0, off # normal width ($D01D &= ~bit0)
sprite expand y 0, on  # double height ($D017 |= bit0)
sprite expand y 0, off # normal height ($D017 &= ~bit0)
sprite priority 0, on  # behind background ($D01B |= bit0)
sprite priority 0, off # in front of background ($D01B &= ~bit0)
var h = sprite_hit()  # sprite-sprite collision ($D01E, cleared on read)
var b = sprite_bg_hit() # sprite-background collision ($D01F, cleared on read)
```


X supports full 9-bit range (0–319): use a **word** variable for runtime values > 255. Sprite data pointer: **data_addr** must be 64-byte aligned; stored as **addr >> 6** at **\$07F8+id**.

Sprite definition

```
sprdef 0
  $00,$3C,$00,  $00,$FF,$00,  $03,$FF,$C0,  $07,$FF,$E0,
  $0F,$FF,$F0,  $0F,$FF,$F0,  $1F,$FF,$F8,  $1F,$FF,$F8,
  $1F,$FF,$F8,  $0F,$FF,$F0,  $0F,$FF,$F0,  $07,$FF,$E0,
  $03,$FF,$C0,  $00,$FF,$00,  $00,$3C,$00,  $00,$00,$00,
  $00,$00,$00,  $00,$00,$00,  $00,$00,$00,  $00,$00,$00,
  $00,$00,$00
end
```

sprdef id ... end embeds 63 sprite bytes at the next 64-byte-aligned address in the code segment, emits a **JMP** over them, and automatically sets **\$07F8+id = data_addr >> 6**. To use the same shape for multiple sprites, read back the pointer:

```
var pg = peek($07F8)  # pointer set by sprdef 0
poke $07F9, pg        # copy to sprites 1-7
```

Memory

```
poke $D020, 2          # STA $D020
poke addr_var, 6        # STA (addr_var),Y - if addr_var is word type
var v = peek($D012)     # LDA $D012
var v = peek(addr_var)  # LDA (addr_var),Y - if addr_var is word type

var w: word = peek16($C000) # read 16-bit little-endian: lo=$C000, hi=$C001
poke16 $0314, $EA81        # write 16-bit little-endian: lo→$0314, hi→$0315
poke16 ptr, w              # word var as address; word var as value
```

peek16(addr) reads two consecutive bytes (lo, hi) as a **word**. **poke16** writes lo then hi.

Disk I/O

```
load "PROGRAM"          # KERNAL LOAD: loads file from device 8 to its native
address
load "DATA", $C000       # loads file to a specific address
load "DATA", ptr         # addr from word variable

save "DATA", $C000, 4096 # KERNAL SAVE from $C000, 4096 bytes → device 8
save "PROG", start, len  # addr and len from word/int variables
```

```

load "PROGRAM"           # KERNAL LOAD: loads file from device 8 to its native
                           address
load "DATA", $C000        # loads file to a specific address
load "DATA", ptr          # addr from word variable

save "DATA", $C000, 4096  # KERNAL SAVE from $C000, 4096 bytes → device 8
save "PROG", start, len   # addr and len from word/int variables

```

`load` calls KERNAL `SETNAM+SETLFS+LOAD` (\$FFBD/\$FFBA/\$FFD5). Without address: secondary address 0 (file's own 2-byte header used as load address). With address: secondary address 1 (file loaded to specified location). `save` calls `SETNAM+SETLFS+SAVE` (\$FFBD/\$FFBA/\$FFD8). Requires both `addr` and `len`.

SID Music

```

load sid "tune.sid"       # embed SID music at its native load address
load sid "tune.sid", $2000 # override: embed at $2000 regardless of SID header

```

`load sid` reads a PSID or RSID file at **compile time**, strips the header, and appends the raw music bytes to the output `.prg`. After `load sid`, two compile-time constants become available:

Constant	Description
<code>sid_init</code>	Init routine address — call once with A = song number (0-based)
<code>sid_play</code>	Play routine address — call every frame (50 Hz PAL) from an IRQ handler

Both constants work anywhere a constant address is accepted: `sys`, `irq`, `poke`, expressions.

Typical usage:

```

load sid "music.sid"

sub music_irq()
    poke $D019, $FF    # ACK VIC raster IRQ
    sys sid_play        # advance one frame of music
    irq_exit            # JMP $EA81: restore A/X/Y + RTI (proper IRQ exit)
end

sys sid_init, 0        # initialise SID chip: A=0 → first sub-tune
irq music_irq, $C0     # raster IRQ at line $C0 → 50 Hz on PAL

sid volume 15          # master volume on

```

Notes:

- SID data is placed **after** all generated code, padded with zeros up to the load address. The compiler warns if the SID load address would overlap generated code.

- PSID v1 and v2 are supported. If the SID header's load address is 0, the first two data bytes are used as the address (PRG-style, little-endian).
- Only one `load sid` per program is meaningful (the last one wins).

Serial channel file I/O

```
open 1, 8, 2, "MYFILE" # open logical file 1, device 8, secondary 2, name
                        "MYFILE"
open 2, 4, 7           # open printer (device 4), no filename
open ch, dev, sec      # channel, device, secondary from variables

print# 1, "HELLO"      # send "HELLO"+CR to logical file 1
print# ch, x, "text"   # any mix of vars, strings – same as print but to file

close 1                # close logical file 1
close ch               # channel from variable
```

`open` calls `SETNAM` (\$FFBD) + `SETLFS` (\$FFBA) + `OPEN` (\$FFC0). Without a filename, `SETNAM` is called with length 0. `print#` routes output via `CHKOUT` (\$FFC9), `CHROUT` per char (+ trailing CR), then `CLRCHN` (\$FFCC). `close` puts the channel number in A and calls `CLOSE` (\$FFC3).

Input

```
input score            # read up to 3 digits from keyboard → 8-bit int var
input "Name: ", name   # optional prompt string, then read line → string var
input "Score: ", score # prompt + int input
```

`input` uses `KERNAL BASIN` (\$FFCF) for blocking, echoed line input with DEL support.

- **Int var:** accepts only 0–9, max 3 chars; converts to 8-bit value on CR.
- **String var:** accepts up to 30 chars; stores as null-terminated string; ZP pair updated.

Float / Fixed-Point

`float` variables use Q8.8 fixed-point format: the high byte is the integer part (0–255) and the low byte is the fractional part (0/256 ... 255/256).

```
var f: float = 3.5      # 3.5 → hi=3, lo=128 (= 0x0380)
var g: float = 0         # integer 0 is promoted to 0.0 automatically

f = 1.5                 # Q8.8 literal assignment
f = f + 1.5             # 16-bit Q8.8 arithmetic (result: 3.0)
f = f + g               # float + float

var n = int(f)          # extract integer part (hi byte) → 8-bit int
print f                 # prints as "N.DD" (e.g. 3.5 → "3.50", 1.25 → "1.25")
```

Operation	Example	Notes
Literal	3.5, 0.25, 1.0	parsed as Q8.8 at compile time
Integer promotion	f = 5	stores 5.0 (hi=5, lo=0)
Add/sub	f + 1.5, f - g	16-bit Q8.8 arithmetic
Extract int	int(f)	returns hi byte as 8-bit int
Print	print f	format "N.DD", always 2 fractional digits

Caveat: Arithmetic overflow wraps at 255.255 (no saturation). Multiplication and division of two float variables are not yet supported — use `int()` + integer arithmetic for those cases.

Math functions

```

var a = abs(x - 20)      # two's-complement absolute value
var b = min(x, 39)       # 8-bit minimum
var c = max(x, 0)        # 8-bit maximum
var s = sgn(score)       # 0 = zero, 1 = positive (1-127), $FF = negative (128-255)
var r = rnd()            # LCG pseudo-random 0-255; seed from raster line
var r = rnd(10)          # LCG pseudo-random 0-9 (rnd() mod n; result 0..n-1)
var s = sin(angle)       # sine: angle 0-255 (full circle), returns 0-255
                        (center=128)
var c = cos(angle)       # cosine = sin(angle+64)

print hex(n)             # print as 2-digit uppercase hex
print bin(n)             # print as 8-bit binary string

```

String functions

```

var n = len(msg)         # length of null-terminated string var (0-255)
var c = asc(msg)         # PETSCII code of first character (0 if empty)
var c = asc("A")         # compile-time: constant PETSCII code
var n = val(s)           # runtime: parse decimal PETSCII string → 8-bit int (e.g.
"042" → 42)
var c = msg[i]           # string character at index i: PETSCII code of msg[i]

```

Number formatting

```

print hex(n)             # print as 2-digit uppercase hex
print bin(n)             # print as 8-bit binary string
print dec(n, 4)          # right-justified decimal in a field of 4 chars (e.g. 42
→ " 42")
print dec(n, width)      # width can also be a variable

```

`dec(n, width)` pads the number on the left with spaces to fill `width` characters. If the number has more digits than `width`, it is printed without padding (no truncation). In non-print contexts `dec(n, w)` evaluates to `n` unchanged (same as `hex/bin`).

REU (RAM Expansion Unit)

```
var ok = reu_present()    # 1 if REU detected, 0 if not (write/read test on $DF04)

reu stash c64addr, bank, reu_addr, len # copy C64 → REU
reu fetch c64addr, bank, reu_addr, len # copy REU → C64
reu swap c64addr, bank, reu_addr, len # swap between C64 and REU
```

`reu_present()` performs a write/read-back test on REU register `$DF04`. Without an REU the write is lost (open bus), so the read-back differs — reliably detects presence without touching any side-effecting command register.

Parameter	Width	Notes
<code>c64addr</code>	16-bit	C64 RAM start — constant, <code>word</code> var, or 8-bit expr
<code>bank</code>	8-bit	REU bank number (0–7 for a 512 KB unit)
<code>reu_addr</code>	16-bit	Offset within the REU bank
<code>len</code>	16-bit	Bytes to transfer (0 = 65 536 in REU hardware)

REU registers: `$DF01` command (`$B0` stash / `$B1` fetch / `$B2` swap), `$DF02–$DF03` C64 addr, `$DF04–$DF05` REU offset, `$DF06` bank, `$DF07–$DF08` length. Transfer is synchronous (CPU halted during DMA). Requires a real REU or VICE: **Settings** → **Hardware** → **RAM Expansion Module**.

Memory utilities

```
fill $0400, 1000, 32    # fill 1000 bytes starting at $0400 with value 32
fill addr, 256, 0        # addr can be word var
fill ptr, len_word, val  # all operands can be expressions / word vars

memcpy $C000, $0400, 256 # copy 256 bytes from $C000 → $0400
memcpy src_ptr, dst_ptr, 40 # word vars for source and destination

drawmem $C000, $0400, 8, 10, 40 # blit 8×10 rect from $C000 → screen at $0400,
stride 40
drawmem src_ptr, dst_ptr, w, h, 40 # word vars for src/dst
```

Both `fill` and `memcpy` support 16-bit lengths (0–65535). Use `word` variables for lengths > 255.

`drawmem src, dst, width, height, stride` copies a 2-D rectangular block. `src` is read linearly (packed rows); `dst` advances by `stride` bytes between rows — use 40 (\$28) for the C64 screen or color RAM (40

columns). Width, height and stride are all 8-bit values. `src` and `dst` may be constants, `word` variables, or 8-bit expressions.

Raster IRQ

```
irq my_handler           # raster IRQ at line 0, handler = sub name or address
irq my_handler, 100      # raster IRQ at raster line 100
irq $C800, 200           # handler at fixed address
irq addr_word            # handler address from a word variable
```

Sets up a raster IRQ via the BASIC soft vector (`$0314/$0315`): disables CIA1 timer IRQ, ACKs pending VIC IRQ, writes raster line to `$D012`, enables VIC raster IRQ (`$D01A=$01`), writes handler address, and re-enables interrupts.

The handler **must** end with `sys $EA81` (KERNAL end-of-IRQ) — plain `RTS` or `RTI` will corrupt the stack. ACK the VIC IRQ first:

```
sub my_handler()
  poke $D019, $FF      # ACK VIC IRQ
  # ... work here ...
  sys $EA81             # JMP to KERNAL end-of-IRQ
end
```

Forward references are supported (`irq my_handler` before the sub is defined).

NMI handler

```
nmi my_nmi              # set NMI vector $0318/$0319 to handler sub or address

sub my_nmi()
  # ... NMI work here ...
  nmi_exit              # JMP $FE47 — proper NMI exit (restores A/X/Y + RTI)
end
```

`nmi handler` writes the handler address to the NMI soft vector (`$0318/$0319`). The hardware NMI vector at `$FFFA` points to the KERNAL NMI routine which branches through `$0318`. The handler **must** end with `nmi_exit` (emits `JMP $FE47`) — using plain `RTI` will corrupt the stack. Forward references supported.

CIA1 timer IRQ

```
cia_timer 19656, my_handler # CIA1 timer A: fires every 19656 cycles (~50 Hz PAL)
cia_timer period, handler   # period can be a variable or expression
```

Sets up CIA1 timer A as a periodic IRQ source via the BASIC soft vector (\$0314/\$0315):

1. SEI — disable interrupts
2. \$DC0D = \$7F — disable all CIA1 IRQs
3. Load 16-bit period lo→\$DC04, hi→\$DC05
4. Write handler address to \$0314/\$0315
5. \$DC0D = \$81 — enable CIA1 timer A IRQ
6. \$DC0E = \$01 — start timer A in continuous mode
7. CLI — re-enable interrupts

The handler must end with `irq_exit` (or `sys $EA81`) and should ACK the CIA1 IRQ:

```
sub my_handler()
    poke $DC0D, $01      # ACK CIA1 timer A IRQ (read also clears it)
    # ... work here ...
    irq_exit             # JMP $EA81: restore A/X/Y + RTI
end
```

PAL timing: clock = 985 248 Hz. Period for 50 Hz = 985 248 / 50 = 19 705 cycles ≈ \$4CC9. Forward references supported.

Compile-time file embedding

```
incbin "sprites.bin"    # embed raw binary bytes at current code position
include "defs.ub"       # inline another .ub source file (lexed+parsed in place)
```

Data / Read

```
data 1, 2, 3, 255       # constant byte table
read varname            # load next byte into varname (auto-declares if needed)
```

All `data` values are collected at compile time. A 2-byte ZP pointer is automatically allocated and initialised at program start. Each `read` advances the pointer.

Inline assembly

```
sys $FFD2               # JSR $FFD2
sys $FFD2, 7            # LDA #7 ; JSR $FFD2 (pass byte value in A register)
irq_exit               # JMP $EA81 — proper IRQ handler exit (restore A/X/Y + RTI)
asm $EA, $EA           # inline raw bytes (NOP NOP) — legacy form
asm {
    ; Full 6502 mnemonics and addressing modes
    LDA #$07           ; immediate
```

```

STA $0286          ; absolute
LDA $50            ; zero-page ($50 ≤ $FF → ZP auto-selected)
STA $0400,X        ; absolute,X indexed
LDA ($50),Y        ; (indirect),Y
LDA ($50,X)        ; (indirect,X)
JSR $FFD2          ; subroutine call
JMP $C000          ; absolute jump
JMP ($FFFC)        ; indirect jump

CLC
ADC #1
SEC
SBC #1

TAX                ; implied / transfer
ASL A              ; accumulator (also just: ASL)
LSR A
ROL
ROR

; Branches – operand is an absolute address; offset is computed automatically
BNE loop           ; forward or backward branch to local label
BEQ done

loop:
  NOP
done:
  RTS

; #<label / #>label – lo / hi byte of a label address
LDA #<handler
STA $0314
LDA #>handler
STA $0315

; * – current assembly location
JMP *

; Raw hex bytes (backward-compatible with old asm { $xx ... } syntax)
$EA $EA           ; two NOP bytes
}

```

Addressing modes:

Syntax	Mode	Bytes	Example
(no operand)	Implied	1	NOP, RTS
A	Accumulator	1	ASL A, LSR
#value	Immediate	2	LDA #\$07
\$zz (0–255)	Zero-page	2	LDA \$50

Syntax	Mode	Bytes	Example
<code>\$zz,X</code>	ZP,X	2	<code>LDA \$50,X</code>
<code>\$zz,Y</code>	ZP,Y	2	<code>LDX \$50,Y</code>
<code>\$xxxx</code>	Absolute	3	<code>LDA \$0400</code>
<code>\$xxxx,X</code>	Absolute,X	3	<code>LDA \$0400,X</code>
<code>\$xxxx,Y</code>	Absolute,Y	3	<code>LDA \$0400,Y</code>
<code>(\$xxxx)</code>	Indirect	3	<code>JMP (\$FFFC)</code>
<code>(\$zz,X)</code>	(Indirect,X)	2	<code>LDA (\$50,X)</code>
<code>(\$zz),Y</code>	(Indirect),Y	2	<code>LDA (\$50),Y</code>
<code>label</code>	Relative	2	<code>BNE label</code> (branches only)

- `$zz` (1–2 hex digits, value ≤ 255) selects zero-page if the instruction supports it; otherwise auto-upgrades to absolute. Use `$00xx` (4 digits) to force absolute.
- Branch operands are absolute addresses; the relative byte offset is computed automatically.
- Local labels (`name:`) are scoped to the `asm { }` block. Forward branches resolved in pass 2.
- `#<label / #>label` yield the lo / hi byte of a label's address.
- `*` yields the current instruction address, so `JMP *` assembles as a self-loop.
- Lines starting with `$`, `%`, or a digit are emitted as raw bytes (backward-compatible).
- Comments: `;` or `//` to end of line. (`#` is the immediate prefix, not a comment.)

Mixing `asm { }` with subroutine parameters

Parameter names are **not accessible** inside `asm { }` blocks. Use UltimateBasic statements to move values into known locations before the `asm { }` block:

```
sub set_colors(border_col, bg_col)
  poke $D020, border_col  # UltimateBasic resolves the ZP address
  poke $D021, bg_col
  asm {
    ; values are already in $D020 / $D021
    LDA $D020
  }
end
```

For routines whose entire body is assembly — especially IRQ handlers that must cross-reference each other — put all handlers in a **single top-level `asm { }` block** in the main program. Labels in the same block share scope, so `irq1` and `irq2` can reference each other freely. See [examples/raster_irq_demo.ub](#).

String ↔ integer

```
numstr score, $0340      # writes "042\0" at $0340 (always 3 digits, zero-padded)
```

```
var n = str_to_int("42") # compile-time: Expr::Number(42)
```

Examples

File	Description
examples/features.ub	const, label/goto, poke/peek, rnd, math functions
examples/new_features.ub	sub params, arrays, word vars, string vars
examples/bitmap_demo.ub	320×200 bitmap, plot, graphics on/off
examples/joystick_demo.ub	joystick reading, sprite movement
examples/mux_demo.ub	raster sprite multiplexer (3 windows × 8 sprites = 24)
examples/orbit_demo.ub	24-sprite orbit with pulsating radius and random colors
examples/plasma_demo.ub	plasma-effect bitmap with raster bar border animation
examples/sprite_data.ub	sprdef shape data (included by other demos)
examples/sprite_mux_orbit.ub	24-sprite orbit demo with sprdef + precomputed positions
examples/sprite_orbit_demo.ub	8 hardware sprites in circular orbit via sin/cos table
examples/reu_bitmap_demo.ub	REU stash/fetch with bitmap graphics
examples/sid_music_demo.ub	SID music player with raster IRQ and keyboard exit

Architecture

```
.ub source
→ Lexer → Vec<Token>
→ Parser → Vec<Stmt>
→ Codegen → Vec<u8> (raw 6502 machine code)
→ mod.rs → PRG = BASIC SYS stub + machine code
```

Two-pass compilation: Pass 1 = main code, Pass 2 = subroutine bodies (appended after main **RTS**). Forward references (**call**, **goto**, **plot**) are patched at the end.

Zero-page layout: **\$02-\$4F** permanent (vars, loop counters, sub params), **\$50-\$7F** scratch (reset per statement), **\$FB** RNG seed.

CLI reference

```
ub build <input.ub> [OPTIONS]
```

```
-o, --output <file>  Output .prg file (default: <input>.prg)
-v, --verbose         Show zero-page layout and code hex dump
--no-stub             Skip the BASIC SYS stub (code loads at $0801)
```

```
--d64 [file]      Also produce a .d64 disk image;
                  without a filename defaults to <output>.d64
-h, --help       Show help
```

After a successful build the compiler always prints a memory map:

```
demo.ub → demo.prg  (386 bytes)
```

```
Load:    $080D - $0989
```

```
Variables (zero page):
```

```
score    ZP:$08    byte
lives    ZP:$0A    byte
msg      ZP:$0C    string
ptr      ZP:$0E    word
```

```
Subroutines:
```

```
greet    $0900
show     $0912
```

```
Arrays ($C000+):
```

```
sprites $C000    8 bytes
```

With `-v` the output additionally shows the internal ZP allocations and a full hex dump.

Building

```
cargo build --release    # binary: target/release/ub
cargo test               # unit + integration tests
```

Known limitations

Feature	Limitation
Integer arithmetic	8-bit unsigned (0–255); <code>word</code> vars hold 16-bit values
Subroutines	No recursion — ZP parameter slots are statically allocated
String vars	Read-only after init; assignment replaces the pointer, not the data
String concat runtime	<code>s1 + s2</code> prints sequentially — no heap allocation or length tracking
<code>rnd()</code> / <code>rnd(n)</code>	Simple LCG, not cryptographic; period = 256
<code>abs()</code> / <code>sgn()</code> / <code>min()</code> / <code>max()</code>	8-bit values only; <code>abs/sgn</code> treat values as signed (bit 7 = negative → <code>abs</code> two's-complements, <code>sgn</code> returns <code>\$FF</code>); <code>min/max</code> are unsigned (0–255)
<code>plot</code>	Out-of-range pixels are silently clipped ($Y \geq 200$ or $X \geq 320$ → no-op)

Feature	Limitation
<code>chr\$</code>	No PETSCII↔ASCII mapping — n is passed as-is to CHROUT
Error reporting	Compile-time only; no runtime error handling